

END TERM EXAMINATION

THIRD SEMESTER [B.TECH.] FEBRUARY 2023

Paper Code: ES-201

Subject: Computational Methods

Time: 3 Hours

Maximum Marks: 75

Note: Attempt any five questions including Q. No. 1 which is compulsory. Select one question from each unit. Assume missing data, if any. Scientific calculator is allowed.

Q. No.	Question (Q1 is Compulsory)	Marks										
1(a)	In the subtraction of 37.593621 - 37.584216 , how many significances will be lost?	[1.5]										
1(b)	How many steps of the bisection algorithm are needed to compute a root of f(x) to full machine single precision on a 32-bit word length computer if a = 16 and b = 17 ?	[1.5]										
1(c)	Using Simpson's ■ rule compute the integral $\int_0^{\pi/2} \sin(x) \, dx$ for n = 4 with an accuracy to five decimal places.	[1.5]										
1(d)	Using Lagrange interpolation find the unique polynomial of degree 2, such that f(0) = 1, f(1) = 3, f(3) = 55 .	[1.5]										
1(e)	Define eigenvalues of a matrix and state fundamental properties of eigenvalues.	[1.5]										
1(f)	Define Cubic Spline function and write the conditions required for cubic spline interpolation.	[1.5]										
1(g)	Solve IVP dy/dx = x + y², y(0) = 1 using Picard's method up to 2nd iterations.	[3]										
1(h)	Classify whether the equations are hyperbolic, elliptical or parabolic: (i) $e^x u_{xx} + \cos(y) u_{xy} - u_{yy} = 0$ (ii) $u_{xx} + u_{yy} + u_x + \sin(x) u_y - u = x^2 + y^2$	[3]										
UNIT-I												
2(a)	If the secant method is used on f(x) = x⁵ + x³ + 3 with x₀ = -1 and x₁ = 1 , what will be x₂ ?	[8]										
2(b)	If f(x) = 3x - cos(x) - 1 and x₀ = 0.6 , what are x₁ and x₂ in Newton iteration?	[7]										
3(a)	[OR] Minimize function f(x) = 4x² + x³ - 7x + 14 within interval [0, 1] using golden section method.	[7.5]										
3(b)	Find minimum value of f(x₁, x₂) = x₁ - x₂ + 2x₁² + 2x₁x₂ + x₂² using steepest descent method taking starting point x₀ = (0, 0) .	[7.5]										
UNIT-II												
4(a)	Prove that e^x = (Δ / E) e^x , the interval of differencing being unity.	[6.5]										
4(b)	Construct a divided difference table for function f(x) given below, find Newton divided difference polynomial, and compute f(0.5) : <table><tr><td>x</td><td>1</td><td>3/2</td><td>0</td><td>2</td></tr><tr><td>f(x)</td><td>3</td><td>13/4</td><td>—</td><td>5/3</td></tr></table>	x	1	3/2	0	2	f(x)	3	13/4	—	5/3	[8.5]
x	1	3/2	0	2								
f(x)	3	13/4	—	5/3								

[P.T.O.]

Q. No.	Question (Continued)	Marks								
5(a)	[OR] If the integral $\int_0^{\pi} e^{\cos x} dx$ is to be computed with absolute error less than $\frac{1}{2} \times 10^{-3}$, and if we are going to use composite trapezoidal rule with uniform partition, how many subintervals are needed?	[6]								
5(b)	Evaluate definite integral $\int_0^1 e^{-x^2} dx$ using basic Simpson's $\blacksquare$ rule and basic Simpson's $\blacksquare$ rule carrying five significant digits. Also find the length of correctness in significant decimal places (rounded) in both rules.	[9]								
UNIT-III										
6(a)	Show that system of equations: $x_1 + 4x_2 + ax_3 = 6, \quad 2x_1 - x_2 + 2ax_3 = 3, \quad ax_1 + 3x_2 + x_3 = 5$ possesses a unique solution when $a \neq 0$, no solution when $a = -1$ and infinitely many solutions when $a = 1$.	[6]								
6(b)	Solve system of linear equations: $0.0001x + y = 1, \quad x + y = 2$ using Gauss elimination with no pivoting and partial pivoting. Analyze how roundoff error affects computation if calculations carry at most five significant digits.	[9]								
7(a)	[OR] Determine the largest eigenvalue and corresponding eigenvector of matrix: $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ using power series method taking starting eigenvector $X^{(0)} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.	[6]								
7(b)										
UNIT-IV										
8(a)	Using Taylor's series method, compute $y(0.2)$ to three decimal precision from $dy/dx = 1 - 2xy$ given $y(0) = 0$. Compare result with exact value of y at 0.2.	[7.5]								
8(b)	Consider initial value problem $dx/dt = 2 + (x - t - 1)^2, \quad x(1) = 2$. Using Runge-Kutta method of 4th order find $x(1.5)$ taking $h = 0.5$.	[7.5]								
9(a)	[OR] Solve IVP $dy/dx = 1 + xy^2, \quad y(0) = 1$ and find $y(0.4)$ using Milne's method, given data: <table><tr><td>x</td><td>0.1</td><td>0.2</td><td>0.3</td></tr><tr><td>y</td><td>1.105</td><td>1.223</td><td>1.355</td></tr></table>	x	0.1	0.2	0.3	y	1.105	1.223	1.355	[6]
x	0.1	0.2	0.3							
y	1.105	1.223	1.355							
9(b)	Solve partial differential equation $u_{xx} + u_{yy} = -81xy, \quad 0 < x < 1, \quad 0 < y < 1$ given boundary conditions $u(x, 0) = 0, \quad u(0, y) = 0, \quad u(1, y) = 100, \quad u(x, 1) = 100$ and $h = \blacksquare$.	[9]								

**** End of Question Paper ****